

Homework Assignment 1

Due date: 1/29

For each of the following data mining tasks, indicate which of the three basic data mining techniques – classification, association rule mining, clustering – can be used:

- (1). An online bookstore has a dataset that contains customer book purchase information. The dataset keeps, for each customer, demographic and socioeconomic information about the customer and the books the customer purchased in the past. You want to know which books are purchased frequently together.
- (2). Assume the same bookstore dataset above. Given the demographic and socioeconomic information of a new customer, you want to know whether the new customer would purchase at least one book from the store.
- (3). Suppose that you have a dataset that has eating habits and physical condition of people. Each person in the dataset has a label indicating whether the person is “obese” or not. You want to use this dataset to tell whether a person is obese or not.
- (4). Assume that you have a dataset which has information about emergency electric generators. For each generator, the dataset has mechanical and electrical properties and CO₂ emission level, whose value is 1, 2, 3, 4, or 5. Given a new emergency electric generator, you want to estimate the CO₂ emission level of the generator using the dataset.
- (5). You have a collection of spam text messages. You hypothesize that each message belongs to one of a few distinct groups. You want to identify these groups and which objects belong to which group.
- (6). Suppose that you have a dataset which has information about people who have/had one or more diseases in the past. From this dataset, you want to analyze the likelihood that a person has/had a certain disease (say heart disease) if that person has/had some other disease (say lung cancer).
- (7). Suppose that you collected a dataset about frogs. For each frog, you have the size, weight, color, habitat, and some other properties of the frog. You want to identify distinct groups of these frogs.

You must justify your answers. For example, you must explain why “classification” should be used for a certain task. If you do not include justification, you will not get the credit for the problem even if your answer is correct.

